

Niels Bohr the Philosopher

Filosoffen Niels Bohr

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Preface

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Chapter X

The Free Will–Determinism Problem

Does the human being have a free will? This is a classic philosophical problem. In more recent times, following the development of mechanical physics in the seventeenth century, it has been sharpened, because physics supplied a full-blooded concept of what it means for something to be determined. From the seventeenth century onward it has been a widespread view that all brain processes are determined. From this one commonly infers that the entire stream of consciousness is likewise determined, and that the experience of freedom of the will is an illusion. Determinism entails that a person cannot himself help his own actions. Take something as simple as a game of chess. One player deliberates for twenty-five minutes over whether to move b2 to b3 or to move b2 to b4. Finally he makes up his mind and moves b2 to b4. According to a determinist it makes no sense to say that he could equally well have moved b2 to b3. Given all the circumstances, he could not have acted otherwise than he did. For his action is determined by all the brain processes that preceded the move and that he himself experienced as deliberations. And these brain processes are in turn determined by others that went before, as well as by his prior bodily state, by the impressions from his surroundings—indeed, by his whole prior life together with his hereditary dispositions. Put differently, he had no choice when he made the move. His move was an inevitable result of everything that went before, just as a grandfather clock's twelve strokes, when the hands stand at twelve, are a result of the clock's mechanical operation. Under the given circumstances the clock could strike nothing other than twelve.

Even the adherents of freedom of the will have been unable to free themselves from the view that all brain processes are determined. They have therefore held that the free will must be understood as a spiritual impulse capable of intervening in the determined course of the brain processes—an impulse that has its origin in something distinct from the material world. This is in itself puzzling, but the free-will philosophers have rightly

been able to point out that full determinism has consequences that even the determinists do not care for. One of them is that concepts such as responsibility, guilt, and duty lose their ethical meaning. Their legal meaning can be upheld, but a human being who is as determined as a clockwork cannot be said, in the ethical sense, to be guilty of what he does, any more than one can say that the grandfather clock is guilty of striking twelve. If the grandfather clock begins to strike only five strokes where it should have struck twelve, one can repair it. Correspondingly one can, perhaps through punishment, “repair” a murderer. But one cannot, in the ethical respect, hold him responsible for what he has done.

Numerous determinists have tried, by all manner of philosophical contortions, to escape this difficulty, but—in my opinion—without success. One of them was Høffding, who believed he could solve the problem by maintaining, with Spinoza, that the assertion that a person is responsible for a committed action is true and has *ethical* meaning provided that all the causes of the action that are essential at the given moment come from the person himself or are, so to speak, located beneath the person’s skin.

This classic discussion between determinists and adherents of freedom of the will Bohr regarded as entirely misconceived. The point of departure for his reflections is the considerations concerning language and our possibilities of description. And the result he thought one could reach fell out neither in favor of the determinists nor of their opponents.

The old problem of free will. Not compatible with determinism, but comparison with indeterminism is equally irrelevant. It is not a question of whether the will is free or not, but must be approached in a different way. The use of the word “will” is indispensable in the account of psychical experiences.¹

In the first place, as noted, he regards a deterministic description of the brain processes bound up with phenomena of consciousness as unrealizable. One might perhaps think that precisely here room was made for the freedom of choice that adherents of freedom of the will have hoped for over the centuries. But Bohr never went along with the idea that “voluntary” actions might be due to unpredictable quantum jumps. These would, among other things, result in wholly random behavior, a so-called *random activity*. Indeterminism leaves as little room as determinism for speaking of personal choice,

¹“Old problem of free will. Incompatible with determinism but comparison with indeterminism equally irrelevant. No question whether the will is free or not, but approach quite different. Use of word ‘volition’ is indispensable in accounting for psychical experience.” *The Karl Taylor Compton Lectures*, M.I.T. MSS, No. 22, 26.8.1957, p. 195. NBA. [Bohr’s original English, as preserved in Favrholt’s note.]

guilt, and responsibility. As little as a person can be held ethically responsible for an action he was fully determined to perform, just as little can he be held responsible for an action that might be due to a quantum jump in a brain cell with an attendant amplifier mechanism.

To make it a bit clearer we may speak of the old problem of freedom of the will. This is a problem that becomes very confusing when we adopt deterministic viewpoints from classical physics, but it is equally irrelevant that there is a display of free will in the indeterminism of quantum mechanics. We are dealing with something quite different, and we do not speak in the way one sometimes does about whether the will is free or not. What we ought to say is: how do we use the word “will.”²

A deterministic description of conscious life is likewise an impossibility. When it is a question of choosing freely, the sharpest examples we can imagine are those in which the choice is preceded by a fully conscious deliberation, as for example in a game of chess. I remember that Bohr once used an example about the decision to buy a house, which was very illustrative.

If I am to tell the story in the first person, it runs as follows: I am deliberating about whether to buy a house. I contact an estate agent, we look at houses together, and finally we find one that appeals to me. Now follows a longer process of deliberation. On the plus side I enumerate the house’s good location, that there is room for the whole family, that it has a large garden, etc. On the minus side I enumerate a number of economic considerations. The monthly expense is at the edge of what I can manage; at some point a major roof repair will have to be undertaken; perhaps in years to come a motorway will be laid right nearby, which will diminish the house’s value, etc. The first thing to notice is that advantages and disadvantages in such a situation will always be incommensurable. Which is better: the location, or the economic security of not buying the house? There is no answer to that. The pluses and minuses are incomparable and so heterogeneous that it makes no sense to speak of being able to logically derive, indeed to deduce, a conclusion—and thereby a decision—from them. But the next point is that the deliberations take place on one plane, while the decision takes place on quite another. *The deliberations can be objectified; the decision cannot.*

Let us say that the estate agent calls and says that now it is urgent, because another buyer has come forward. I must make up my mind today. I go through all the deliberations

² *The Karl Taylor Compton Lectures*, M.I.T. MSS, No. 22, 26.8.1957, 5th Lecture, pp. 12–13. NBA.

once more. Then I decide to buy the house. While the deliberations were going on, I could write them all down, discuss them with others, set them aside, take them up again, etc. That is to say, in the deliberation phase I could place a divide between the deliberations and the subject. In deciding, there is no divide between the subject and the act of decision; I am, so to speak, myself the decision. This shows itself in the fact that the decision cannot be described. I could, if I liked, write a whole—rather boring—book about the house purchase and fill chapter after chapter with my deliberations *pro et contra*. But the chapter on the decision itself could amount to no more than: Then I made the decision.

The decision itself annuls the deliberations. Once the decision has been made, they are no longer deliberations for me. They have become something purely historical. It would be absurd if someone said to me: “Hurry up and make that decision, so that you can always deliberate afterwards.” That would be an impossibility, because a deliberation is a deliberation only so long as the decision has not been made. What happens in the decision is that the divide between the subject and the observed content of consciousness (deliberations, and the like) is suddenly abolished. Since a causal description is feasible only if one can maintain a clear divide between subject and object all the way through, the decision cannot, in the psychological respect, be explained as an effect of the preceding deliberations. In the light of this account one may read the following quotations:

... the feeling of will is given its natural scope, since we, in the situations in which we use the word “will,” on account of the nature of the situation do not distinguish between subject and object in a way that permits us to use the idea of causality. Such a sentence may seem complicated, but I think it is a simple way—against the background of the images from atomic physics that we picture to ourselves—of attaining a very precise description of the situation.³

If one wishes to investigate some psychological experience or other, this means that one must sort out the different elements, and attention must be fixed on certain aspects of the matter. But this is very difficult, for if one thinks of some experience and fixes one’s attention on different aspects of this experience, then it is far from being the same thing—it comes to concern two entirely different psychological experiences. This is so well known that it has entered into language ... *The situations in which we use the expression “I will” are situations in which there is no question at all of making*

³Hitchcock Lectures, March 1937, p. 353. NBA.

such a distinction between a subject and an object as could at all provide a background, the first necessary presupposition for a causal description [My italics, DF]. These are in themselves things that psychology knows well, but in this way the concept of will and the feeling of will are given their scope.⁴

Here the decisive point is that the cognition of a state of consciousness in which words like “I will” find application is complementary to a state in which we occupy ourselves with an analysis of the motives for our actions. It is against this background that we can, in a rational manner, speak of will as an indispensable element in any attempt to give an exhaustive description of consciousness.⁵

If we speak of a situation in which we have, so to speak, a feeling of will, then such a situation cannot be compared with, but will exclude, a situation in which we reflect on the purposes of our actions. This can be described by saying that we are dealing with things that exclude one another, because in a way we are speaking about ourselves, but which have a different dividing line between what is the content of consciousness on which we focus and the background that we loosely refer to as ourselves.⁶

Since we human beings throughout the day make decisions or effect choices (my own sense of the difference between a decision and a choice is that decisions occur after thorough deliberation, while choices are made spontaneously or after a brief deliberation—but Bohr says nothing about this), this means that the stream of consciousness cannot be conceived as an inner film in which one image succeeds another. That is the view one may find in the English empiricists, especially David Hume. In him the content of consciousness is described as a succession of “ideas,” that is, more or less mutually separate representations that succeed one another like images on a cinema screen. Sense impressions too, according to Hume, are such separate units following one upon another. But according to Bohr it is likewise no use to characterize the stream of consciousness as a continuous stream of phenomena that engage or overlap one another. This too is a misconception. What is epistemologically decisive is that, in order to be able to speak and act at all, we must continually shift the dividing line between the subject and the rest of the content of consciousness.

... whereas at the early stages of physics one could rely directly on those

⁴*Science and Life*, MSS, No. 16, 1939, p. 13. NBA.

⁵*Medical Research and Natural Philosophy*, 1951, p. 971.

⁶*The Karl Taylor Compton Lectures*, M.I.T. MSS, No. 22, 26.8.1957, 5th Lecture, pp. 12–13. NBA.

events in daily life that permitted a simple causal account, from the very beginning of languages one has employed an essentially complementary description of our content of consciousness. The use of language adapted to such communications points, after all, not to an unbroken course of events, but rather to separate, mutually exclusive experiences characterized by a different placement of the dividing line between the content on which attention is directed and the background hinted at by the expression “ourselves.” / A particularly striking example is the relation between situations in which we consider the motives for our actions and situations in which we experience a feeling of will.⁷

It is this relation that makes it possible for us in daily life to distinguish between situations in which one “did something on purpose” and others in which one “could not help it.”

Bohr held, as said, that the traditional discussion of free will versus determinism was misconceived. What he himself thought he could contribute was not a solution to the traditional problem, but rather a kind of explanation of why, from the very beginning of language, one must necessarily have the distinction between “doing something on purpose” and “doing something one could not help.” And he had no intention of developing a grand philosophy of freedom of the will. It may strike the reader as irritating that Bohr, in all the lectures where he touches on the problem of free will—and there are many—always confines himself to a very few lines on the problem, even though he speaks of the problem as immensely interesting and returns to it again and again. Léon Rosenfeld says of this:

I think he has published practically everything. He always insisted that “It is only one sentence, you see.” It was never gone into more closely without his insisting that this ought not to be done. He said of freedom of the will, “I have nothing more to say about freedom of the will than the few sentences I have written down.” He could only spoil it by saying more about it, because it was the essential epistemological aspect, and there was no more to say about it, because those words were used in an unanalyzed way; a concept such as freedom in connection with will was a primitive concept according to him, and therefore there was no more to say. One does not discuss the meaning of the whole numbers, for they are the beginning of mathematics; there is

⁷*Kundskabens Enhed* (1954), AME, pp. 93–94.

no more to say about it. That was the attitude he had; this epistemology was not to be a pretext for rambling twaddle by which one could only render the subject obscure. Rather, it was to point out that one uses these primitive concepts in different situations, and that they cover the situation adequately.⁸

The comparison with the whole, positive numbers in this quotation is interesting. As little as it makes any sense to analyze the whole numbers (there is no agreement on this, but I am reporting Bohr's position), so little does it make sense to say anything more precise about will—that is, to attempt to answer such questions as: What really is will? What one must establish is that, inasmuch as we can have cognition, we must have consciousness, and inasmuch as we have consciousness, we have the possibility of reflecting on our own states of consciousness. And inasmuch as we can do this, we can shift the boundary between subject and object through self-observation.

In my opinion Bohr must have meant something roughly like this: When a child learns language, it simultaneously learns to distinguish between what is subjective and what is objective in the ordinary everyday sense. It learns that dreams are not real. It learns to speak of chairs and tables without reference to how it experiences them. It also learns to speak of feelings and sensations and thereby learns that these are first-person experiences that are, phenomenologically speaking, private. It also learns to distinguish between when it does something on purpose and when it cannot help what it happened to do, precisely on the basis of an unconscious understanding of the first-person observational situation. And it learns quite early to master these concepts, so that it is able to tell others what it does on purpose and what it happened to do without willing it. It learns to understand turns of phrase such as “He did it on purpose,” “He chose to keep silent,” “He could not help coughing,” etc. But all of what is here mentioned is learned, so to speak, *en bloc*. One cannot learn to speak unambiguously without learning the difference between “subjective” and “objective” and “public” and “private,” and thereby also the correct use of “cause” and “free will.” “Before,” “after,” and “now,” “here” and “there,” and “thing” and “whole numbers” are other concepts that enter into this “fundamental language” that constitutes the core of everyday language. To learn these concepts does not mean arriving at being able to define them. Here we must distinguish between “knowing how” and “knowing that” (a distinction found in G. Ryle's *The Concept of Mind*). For example, the great majority of Danes master the use of the present, past, and future tenses, the subjunctive, the conjugation of verbs, etc. That is “knowing how.” But only very few would be able to write a Danish grammar and, for

⁸Rosenfeld interview. NBA.

instance, be able to formulate the rules for when to use the subjunctive, for that requires a “knowing that.” A child can perfectly well master the subjective/objective distinction without knowing these words and without being able to define them. If we assume that the emergence of language in a distant prehistoric time took place in a social interplay in which individuals were able to communicate to one another about the things in their surroundings and to distinguish between their private experiences and their shared experiences, then Bohr may be right, as he puts it, that the concepts “cause” and “free will” are as old as language itself. And that the complementarity between thought and feeling was known to the first users of language. Some quotations to illuminate this:

The word will necessary for the description of psychical experiences, indispensable in practice. 2. The question, however, is in what sense we can say that we determine our actions. 3. All talk of freedom in this respect is excluded if we assume deterministic predestination.⁹

Even if we wish to describe will as a subjective sensation, we are in this way led to regard the concept of freedom of the will as an inherent part of an objective description of psychical experience.¹⁰

The demand for causality is an idealized practical demand from daily life, corresponding to the wish to survey the consequences of actions. Its immediate limitation in daily life is our own and others’ volitional actions—a question that is kept entirely outside classical theory and finds its place in Descartes’ distinction between soul and body.¹¹

... to recall that the application of the word will in the objective description of our situation corresponds entirely to such words—equally indispensable for human communication—as hope and responsibility.¹²

The point is that the will is felt to be free, and that this is a use of words that we cannot elaborate further, since each word is defined only by the context, and the context in this case is just as objective a mode of description as in any other domain.¹³

The language we use in human intercourse to communicate our state of mind is, after all, very different from that ordinarily used in the physical

⁹*Atomerne og den menneskelige erkendelse*, MSS, No. 21, 16.4.1956. NBA.

¹⁰*Physical Science and the Study of Religions*, MSS, No. 20, 18.10.1953. NBA.

¹¹*Unity of Knowledge*, MSS, No. 21, 6.5.1954. NBA.

¹²*Atomerne og den menneskelige erkendelse* (1955), AME, p. 112.

¹³*Physical Science and the Study of Religions*, MSS No. 20, 19.9.53.—It says 52, but that must be an error, DF. NBA.

sciences. Thus words like “thinking” and “willing,” which refer to situations that mutually exclude one another but that are equally characteristic of conscious life, have been used in a typically complementary manner ever since the very origin of language.¹⁴

It is one thing that one cannot analyze one’s own content of consciousness in such a way that it can be fitted into the framework of a causal description. But how do matters stand if one attempts to describe another person as a deterministic system? Can one in principle predict the behavior of others?

Bohr’s view is that this will never be possible. In the first place, it is inconceivable that, on the basis of a physiological knowledge of another person, one could obtain a basis for predictions. The uncertainty relations set a limit to the description of the processes in the brain and central nervous system, or put differently: what happens in the brain is not processes in the classical physical sense, and a brain state is not a “state” in the classical physical sense. But could one not imagine that, on the basis of a thorough knowledge of another person, one could predict what he would do in a given situation in which, in accordance with normal usage, we speak of his choosing freely? We can of course predict that a person will pull his hand back if he inadvertently comes to rest it on a red-hot hotplate, but would we be able to predict what move he will make in a given situation in a game of chess, or choose to do in similar situations involving well-considered decisions? Bohr writes of this:

The decisive point is that, if we try to predict what another person will decide to do in a given situation, we must not only explore his whole background, including all the aspects of his life history that may have contributed to forming his character, but we must understand that in the last analysis we are striving to put ourselves in his place. It is of course impossible to say whether a person wishes to do something because he believes he can, or whether he can because he wills it ... in this sense we can, both practically and logically, speak of “the freedom of our will” in a manner that gives the requisite scope for the use of words like responsibility and hope, which, taken individually, are as little definable as other words indispensable for human communication.¹⁵

As mentioned earlier in this book, the very expression “to put oneself in another’s place” has a number of different meanings, depending on the situation one is speaking

¹⁴*Forbindelsen mellem videnskaberne* (1960), AME II, pp. 34–35.

¹⁵*Kundskabens Enhed* (1954), AME, pp. 95–96.

of. If a person says that he froze and stood stock-still when he suddenly found himself facing a bear in a Canadian wilderness, another can quite meaningfully say, "In your place I would have run to the car and jumped into it." If a person has a headache but refuses to take a painkiller, it also makes a certain sense to say, "In your place I would take it." But if a person suffers from paranoia and believes himself persecuted by a secret police that hides behind every tree and every corner, then it makes little sense to say, "In your place I would pull myself together and say that it is all imagination." For the other can with full right reply, "In my place—that is to say, if you were me—you would be just as afraid as I am. If you claim that you can enter into my situation without feeling this fear, then you have not entered into it."

We cannot predict what another person will do in a situation in which there are many possibilities to choose among. Even if we know his entire past and everything he has thought, we cannot fully and wholly put ourselves in his place. But this means that we cannot establish a deterministic third-person description. And note: if we could put ourselves in his place, then we would find ourselves in a first-person description situation, and we (or I should perhaps say I, if it is I who empathize with his situation) would be as little able to predict his decision as he himself is—or as we ourselves can predict our own decisions. For I cannot predict my decisions. If I could, they would precisely not be decisions

—or else they would be made in the very act of my predicting them. If, for example, I predict with *100 percent certainty* that in ten minutes I will decide to drink a cup of coffee, then my action in ten minutes will not be the expression of a decision at that particular time. So the very expression "in ten minutes I will decide to drink a cup of coffee, and I know it with 100 percent certainty" is nonsense. It can, on the other hand, be understood as a clumsy expression of the fact that I have already now decided to drink a cup of coffee in ten minutes. But this decision I could not predict either.

Since Niels Bohr's time many philosophical treatises have been written on the free-will problem without our having come any closer to a final solution; there are "schools" that "wage war" on one another, and in philosophical survey works one can read over and over again what these "schools" amount to, with all the usual labels: determinism, indeterminism, necessitarianism, libertarianism, compatibilism, etc. Here I will not go into all this, but will instead pose the problem of whether a general ontological determinism is at all compatible with the concept of cognition. If everything in the universe, including all human behavior and conscious life, is determined, is cognition then at all possible? It was this problem that I wrote a little treatise about in my student

days, and which Bohr read and declared himself in agreement with.

But before I discuss the treatise's main argument, I will briefly recount and criticize a contribution on freedom of the will—or rather the lack of it—that has appeared in the debate of recent years. I am thinking here of Benjamin Libet's experiments.¹⁶ Experiments that many now believe to be the final solution to the two-thousand-year-old problem.

Libet conducted a series of experiments in which, using various forms of brain scanning, he sought to determine the temporal relation between the brain impulse that precedes a volitional action and the experience one has of deciding to carry out the action. In Danish we have a very vivid description of the experiments in Tor Nørretranders' in many ways thought-provoking book *Mærk verden*.¹⁷ According to Libet, neurological research has shown that all volitional actions are initiated by the brain wholly independently of any conscious act of will. He thought he could establish that when one decides to carry out a willed movement, for example a movement of the hand, the cerebral impulse occurs about 500 ms (thousandths of a second) before the impulse reaches the muscular system. The feeling of an intention, on the other hand—the experience of the wish to move the hand—sets in only 150 ms before the movement is carried out. Libet believes that with his experimental results he can overturn the traditional conception of what takes place in a volitional action.

Throughout the thousand-year-old debate there have been extreme viewpoints concerning the so-called mind-body problem. Adherents of free will have, within this framework, maintained that a person can in one way or another decide on something, for example to clench his right hand, and that the decision is conscious for the person, either before the action is carried out or as it is carried out. The proponents of universal determinism have been divided broadly into two camps. In the one camp are the adherents of *epiphenomenalism*. This holds, as mentioned, that although there are phenomena of consciousness—experiences, memory images, decisions, and the like—they are in reality nothing but a kind of by-product of the brain processes, a kind of after-echo that tells a person what his brain and body are up to. In the other camp we find the philosophers who hold that consciousness is an independent “mode of existence” alongside—but coupled in an inexplicable way with—the functions of the brain.

Libet's point of departure is an epiphenomenalist conception. And he believes that his experiments prove that our concept of free will is an illusion. It is, so to speak, not we who make the decisions, but the decisions that make us. When I stand in the cafeteria

¹⁶Benjamin Libet: 'Unconscious cerebral initiative and the role of conscious will in voluntary action'. *The Behavioral and Brain Sciences* 8, 1985.

¹⁷Tor Nørretranders: *Mærk Verden*. Gyldendal, København, 1991.

at the university and think, “Shall I now choose tea or coffee?” and then choose coffee, I experience that I made a free choice before my hands chose a coffee cup and turned on the coffee machine. But according to Libet, my brain—without my knowing it—had made the decision 350 milliseconds, or 3.5 tenths of a second, before I experienced that I myself, that is, my conscious I, made the decision. And my experience of the decision set in 1.5 tenths of a second before the action. So, ergo, it was not my conscious I that made the decision.

It should just be mentioned that John-Dylan Haynes has recently repeated Libet’s experiments and has arrived at the finding that the time interval between the brain process correlated with the decision to press a button—after a prior instruction to carry out this action at some time or other—and the consciousness of carrying out the decision can be as much as seven seconds.¹⁸ So it is not by virtue of a conscious act of will that the test subject carries out the action. The experience that it is voluntary comes later than the brain has “decided” to carry out the action, and is therefore an illusion. Expressed in the first person singular, the matter must be that it is not I who make the decision, but the decision that makes me. So Haynes thinks.

And Haynes concludes that it is hereby finally proven that the human being has no free will. All our actions are causally determined; we must confess to determinism: everything we say, think, and do is unambiguously determined by antecedent causes. When one holds a human being responsible for an action, one presupposes that the person could have acted otherwise than he or she did. But that is not the case. No one can think, hold an opinion, or act otherwise than they do, so concepts such as responsibility, guilt, and duty are hereby emptied of their ethical content.

Libet’s view has been criticized—rightly, in my opinion—by Bennett and Hacker in the book *Philosophical Foundations of Neuroscience*.¹⁹

They argue that it is neither a necessary nor a sufficient condition for the performance of a volitional action that a feeling of desire or intention should precede the action. Throughout the day one performs a multitude of volitional actions in which no experience of purpose or intention occurs before the action. They give a couple of examples: When one is about to write a letter, one automatically reaches for the pen—there is, after all, no prior act of consciousness. If the doorbell rings, I put down my book, rise from the armchair, and go and open the door—or else I think, “It’s probably a couple of missionaries,” and remain seated. But if I go and open the door, the process is not something that implies that I experience anything like “I had better get up,” “I had better

¹⁸*New Scientist*, 19 April 2008.

¹⁹M.R. Bennett & P.M.S. Hacker: *Philosophical Foundations of Neuroscience*. Blackwell, Oxford, 2003.

go and open the door.” It may be that I am thinking of something entirely different in the time from the doorbell’s ringing until I open it, for example, “Oh, how stiff-legged I am today, drank too much red wine yesterday, must get some more exercise,” and that means that the decision to open the door is unconscious, despite the fact that the action is a volitional action.

Throughout the day a normal person performs a multitude of actions that are not preceded by any conscious decision, and makes a multitude of decisions that are not immediately followed by an action. Examples of the latter we all know: Oh, there’s no more sugar, I must remember to buy some tomorrow; Goodness, I have completely forgotten that Celine has her birthday on Thursday, I must remember to send her a card, etc. And as for the former, there is an immense complexity in everything we say and do. And what many overlook is that a normal person, with ordinary circumspection and in full possession of his senses, has an incredibly fine-meshed appraisal of what are volitional actions and what are unpremeditated, of what are intended actions and which are spontaneous, in other people’s behavior and speech.

Take, for example, something like a football match, where many free kicks and some penalty kicks are awarded. In each individual case the referee must decide whether it is a matter of unintentional occurrences or intentional actions, whether the player at whose expense it goes in a given situation is “diving,” and so on. When one watches the various situations in slow motion on television, one has a sure and correct assessment in about 80–90 percent of cases. Generally the players, when they are penalized, acknowledge that they committed the fouls and that they acted deliberately, but not that any conscious decision in the form of some phenomenon of consciousness or other preceded the action, for that rarely happens.

Or take something like an ordinary conversation. I am speaking with someone about his impending divorce. With the intention of instilling in him a little optimism regarding the future, I choose my words with care. I do not say “it will all work out,” for that is too superficial a remark, so I say something else—“the first period is a little hard, but when you look back a year from now . . .” and so on. But what I say, I choose while I am speaking, without prior decisions, though of course with an intention that I now and then register, but which is not a special state of consciousness that is “on” the whole time.

There is a quite different aspect of the free-will problematic that Benjamin Libet has not caught sight of, and that Bennett and Hacker mention just in passing but make no great point of. It concerns decisions that one makes after a longer period of deliberation,

and that are therefore of a different kind from the decision to clench one's right hand. Here I am thinking of the above-mentioned example about a house purchase, which Niels Bohr put forward in a discussion of freedom of the will.

If one is about to buy a house, Bohr said, then there can be a long course of events. One contacts an estate agent, looks at various offers, is attracted to one of them over the others, and begins to consider what pluses and minuses there may be in connection with a possible purchase. All the pluses and minuses one can write up and discuss with others, but the interesting thing is that they are incomparable, incommensurable, so they do not lead with logical necessity to any conclusion. Is district heating to be preferred over the traffic problem? If the pluses and minuses could lead to a conclusion with logical necessity, then it would indeed be conceivable that Libet could be right that the brain had made the decision before the buyer made his. But now, in Bohr's example, what happens is that the estate agent calls, another possible buyer has intervened, so now our hero must make a decision, and he does. He strikes and buys the house.

Niels Bohr, for good reasons, knew nothing of Libet's argumentation. What interested him was the mutual exclusion relation between the deliberations and the decision, for at the moment the decision is made, the deliberations are, as mentioned, no longer deliberations, but annulled. And what occupied him was also that a decision, in contrast to the deliberations, does not exist in time, is not something one can analyze, is not a process, but has a closedness such as we know from quantum phenomena. But what is at issue here is whether one can maintain belief in determinism in connection with a decision that is made after a series of deliberations. Here Libet's argumentation is irrelevant. Whether the brain reacts before the subject's experience is subordinate. The question is whether all the decisions that a human being makes throughout life can be described as long causal chains in which the individual has, so to speak, no co-determination. And the next question is: if one says yes to the first question, how then will one demonstrate that this is the case? Purely methodologically it would indeed require a lifelong ongoing interview of an individual, supplemented by everything that happens in his immediate surroundings, plus an ongoing registration of his brain activity; and as regards the last, the deterministic description would soon fall short, since a good deal of brain activity takes place at the quantum-mechanical level.

Bennett and Hacker also argue that one's consciousness of what one will do a few seconds hence cannot be adduced as a sufficient ground for an action's being volitional. I can, for example, feel that I am about to sneeze, and after a few seconds I then sneeze; but that does not make the action volitional.

As said, Libet registers the brain impulse that precedes one's moving one's hand "on purpose," and thereafter the test subject's experience of the act of will, which according to Libet occurs 350 ms later than the brain impulse. But how can he know when the test subject has this experience? Here he uses a modernized version of the so-called complication clock, invented by Wilhelm Wundt (1832–1920). As it happens, I worked a great deal with this complication clock when I studied psychology at the University of Copenhagen in the 1950s. We had at that time so-called laboratory exercises with the subsequent preparation of experimental reports, and one of the exercises involved the use of the complication clock. This consists of a large clock face marked with the numbers 1, 2, 3, 4, and so on, as on an ordinary clock, but with only a single hand, which can be set to rotate one full revolution in, for example, 1, 2, or 3 seconds. The experiment was designed to show how one could advance or delay the processes in the auditory nerve relative to the processes in the optic nerve. One could set the complication clock to give a clicking sound when the hand passed the number 5, but if one, for example, set oneself for the click to appear and be heard only when the hand passed the number 6, one could actually make it fit with what one experienced visually, because one's set delayed the process in the auditory nerves and thereby the experience of the click. As a rule there were two students to carry out the many different experiments, because one could then be experimenter and test subject by turns. My partner was a female student, Barbara Knox. We discovered that one's experience of the hand's speed and positions depended on whether one fixated—that is, directed one's gaze toward—the center of the face, or focused on points to the right or left of the face or above and below it. The distance to the face also played a role, and since no one had discovered these things before, we were allowed, with keys handed out to us, to use the laboratory for a whole autumn holiday for our investigations.

The result of our work was that the complication clock was a poor instrument for dating experiences in time. And in Libet's experiments it is not the most fortunate method, even though Libet has replaced the clock face and hand with a television screen on which the points corresponding to the clock's 1, 2, 3, 4, and so on are replaced by spots of light that are switched on and off serially. When it is a matter of milliseconds, there are too many possibilities of error in the perception, to which is added the test subject's reaction speed, so for the time being there is no reason to believe in the results Libet thinks he has arrived at. But even if he should be right, the idea of universal determinism is more than hard to come to terms with.

Laplace speaks, as previously mentioned, of a universal spirit that can cognize all

forces and the position of all things in the universe at a given moment, but the thought experiment is a play on words. For, as said, it holds that either the universal spirit is *outside* the universe, whatever that might mean, or else he is *inside* the universe. If he is outside, he cannot receive any information from the universe: he cannot see the placement of things or register the acting forces, for that would have to mean that he received information in the form of light waves, radio waves, or the like, and that simply cannot be done. But if we assume that he is *in* the universe, then the problem is that he can indeed have a comprehensive quantity of observations, but cannot observe what goes on within himself. And as a part of the universe he is subject to the observational conditions that we meet in quantum physics, which, as mentioned earlier, cannot be fitted into a deterministic description.

There are still physicists who believe that Laplace's thought experiment can be revived in a present-day physical formulation. Since one can ascribe to all particles a Schrödinger wave equation, and since one can construct such wave equations for ensembles of particles in any arbitrary quantity, there must in principle be able to be formulated a—deterministic—wave equation for the whole universe. But again: where does the observer stand, the universal world-spirit who is supposed to attain the all-encompassing cognition? The argument against this view runs along the same lines as the argument against Laplace.

But there is yet another argument against general ontological determinism. And it is precisely the one that I wrote to Bohr about many years ago, and which he went along with. Suppose that determinism were true, valid, correct. It would then have to hold that all our brain processes are determined and, according to epiphenomenalism, that all our phenomena of consciousness were determined by the brain-physiological conditions. But if this were the case, then our brain processes and phenomena of consciousness would, in both the ontological and the epistemological respect, have to be put on a par with natural phenomena in general. But of natural phenomena such as thunderstorms and waterfalls it holds that they cannot mean anything. A thunderstorm can flash and rumble, but not put forward a viewpoint. A waterfall cannot assert anything or hold anything to be true. A downpour can express neither deliberation nor doubt. And the same would then have to hold—if general ontological determinism were correct—for the human being. No human being would be able to hold anything to be true and argue for it. No human being would be able to mean anything. A person's opinion would lie on a par with his pulse and his sweat secretion. No one would be able to hold determinism to be true, for if he did, and others praised him for his philosophy, it would—if determinism

were correct—merely be a product of air vibrations and printer’s ink, which could have been otherwise if other causal processes had lain behind it. If the adherent of free will and an adherent of determinism came to quarrel, neither of them would be right. For if determinism were valid, they would both be determined to hold to their respective positions, and there would be no “court of philosophy” to decide who was right—as little as one can decide whether the rumbling of the thunderstorm or the splashing of the rain stands for any truth or cognition.

That so few philosophers through the ages have been attentive to this is connected with the fact that the discussion of determinism versus free will almost always focuses on human actions. “He acted thus and so, but is he guilty if he could not have acted otherwise?” is how it goes. I have in this connection no great prophets to whom I can refer (apart from Kant—with certain reservations). But my thoughts concerning this probably go back to K. Grue-Sørensen’s *Studier over Refleksivitet*.²⁰

Grue-Sørensen writes that if you hold general determinism to be true, you must necessarily assume that you are not determined to hold it to be true. For if you are determined to hold it to be true, you must accept that the one who holds it to be false is also determined to do so. But then you have no argument in favor of your own view over against his. You are not free to maintain that you are right.

Here Grue-Sørensen²¹ refers to an argument in Heinrich Rickert (1863–1936), who states that the one who advocates determinism removes himself from a passive contemplation of the states of affairs. For he asserts determinism and denies indeterminism, and “Da muss sich sofort die Frage erheben: ist er bei dieser Stellungnahme frei oder nicht?” [“Here the question immediately arises whether this taking of a position is free or not,” DF]. If the determinist answers this question by saying “that he is unfree and causally determined to assert his view,” then he has at the same time lost every advantage over the indeterminist, who is in exactly the same situation. And if it is answered by saying that in taking a position for determinism he was free, that is, independent of the causal compulsion, then in this answer he has contradicted his own doctrine. This is Rickert’s construal of the problem. Now one may surely assume, Grue-Sørensen writes,

that every determinist will opt for the first part of the alternative and assert that his assertion of determinism is of course causally determined like everything else, but that this serves as no argument against the doctrine. It is also correct that there is nothing whatever in it that implies that determinism

²⁰K. Grue-Sørensen: *Studier over refleksivitet*. J. H. Schultz Forlag, København, 1950.

²¹p. 191.

should be false. It has only, by its content, forfeited its advantage over the opposite view. This in turn rests on the fact that, according to this view, our thinking is to such a degree fixed in one context, the causal, that there comes to be no mobility in the other, the logical; and this mobility under the aspect of correctness is presupposed in all our thinking and arguing against one another.

I prefer to express the matter in this way: that we are compelled to assume that our thinking is to a certain extent not determined, if we are to be able to apply the concepts “truth,” “falsity,” and “meaning” in an intelligible way. It is a presupposition for being able to think that one can, within certain limits, freely choose between different attitudes or points of view, and that one can, so to speak, freely choose what one will think about in the next second. And I believe I can establish that Niels Bohr was in agreement with me—and that is to say, with Grue-Sørensen.

Chapter XI

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